

AI, Energy, & Water: Predictions?

Posted by u/jewishagnostic • 0 points • 9 comments

There's a lot of discussion about the environmental impacts of AI, particularly its energy and water needs.

I kind of expect that r&d on more efficient computers will reduce those demands over the next few years, but I tend to be a bit optimistic.

I think we'll still need to expand our energy supplies since even if AIs become 1000x more efficient, demand for AI can easily grow 10,000x, overwhelming the efficiency gains. Still, I obviously see this as more manageable and sustainable than the current systems.

Also, I expect major AI companies to continue to invest in creating sustainable energy sources.

And I even suspect that the new generation of water-from-air devices [e.g. MOF based], will allow data centers to harvest water from air, thereby reducing the stress on local water supplies. So even if they require a lot of water, they'll just 'create' it themselves from the air.

What do you all think? What are your expectations?

Comments

u/Ri711 • 2 points • 2025-01-23 03:22 UTC

Great insights! I agree, while AI efficiency will improve, the demand for it could still outpace those gains. Expanding energy supplies and investing in sustainable energy is key. The idea of using water-from-air devices for data centers is also exciting and could help reduce local water stress. I recently read a blog that covers the topic of the future of sustainable living, and it really got me thinking about how these advancements will shape our future. Your optimism is encouraging, and innovation will play a big role in tackling these challenges.

u/Artemistical • 1 points • 2025-01-21 14:18 UTC

any recs of a good read to learn more about how AI uses water/energy in order to work?

u/CareerAdvised • 1 points • 2025-01-21 01:47 UTC

If that helps you sleep better: I'm working on a concept for productive, reproducible, long term sustainable, regenerative agriculture with AI.

So far, AI has delivered a great deal of value.

The investment is a purely human responsibility and humanity is falling short.

u/blimpyway • 1 points • 2025-01-20 20:56 UTC

If AI becomes 1000 times more efficient and demand increases 10000 times, it means we need only 10 times the resources to match a 10000 times greater demand.

Quite a bargain I think.

u/jewishagnostic • 1 points • 2025-01-20 22:08 UTC

agreed. but we still have to manage our resources. it's a bit like that situation where (so I've heard) people in oprah's audience who got big gifts like free cars found themselves in trouble when they didn't have the liquid cash to pay the necessary taxes. even the best deal (free!) must be managed with existing resources.

u/blimpyway • 1 points • 2025-01-20 22:19 UTC

>but we still have to manage our resources

That should be covered in the 10000x demand increase of AI

u/jewishagnostic • 1 points • 2025-01-20 22:22 UTC

unfortunately, not necessarily. ai is still led by capitalism

u/critiqueextension • 1 points • 2025-01-20 20:24 UTC

The environmental impacts of AI, especially in terms of energy and water usage, are more severe than often expressed, with significant increases in both electricity and water needs for data centers revealed by recent studies. While advancements like water-from-air technology are promising, the reality is that AI models currently consume thousands of liters of water per training session, raising concerns over sustainability amid their increasing adoption in business and technology sectors.

* [Explained: Generative AI's environmental impact](<https://news.mit.edu/2025/explained-generative-ai-environmental-impact-0117>)

* [As generative AI asks for more power, data centers seek ...]

(<https://www2.deloitte.com/us/en/insights/industry/technology/technology-media-and-telecom-predictions/2025/genai-power-consumption-creates-need-for-more-sustainable-data-centers.html>)

* [AI's Impact on the Environment, Explained](<https://www.snopes.com/news/2025/01/16/ai-environment-carbon-footprint/>)

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u/Asclepius555 • 1 points • 2025-01-20 17:17 UTC

I see this as one of those demands we will justify as easily as we now justify the resources used to let us consume other animals like pigs and cows.

u/hype-deflator • 3 points • 2025-01-20 17:12 UTC

I predict that AI will balance everything out, and everyone will live happily ever after.

I mean, if the sea levels are rising and the AIs drink up all the extra seawater we're good, right?

u/jewishagnostic • 2 points • 2025-01-20 17:15 UTC

